

Chapter 16.3: Conservative Vector Fields

V2: Conservative Vector Fields. I can test for conservative vector fields and find potential functions. I can state and apply the Fundamental Theorem of Line Integrals.

Mechanics

1. Show that the vector field $\mathbf{F} = 12xy\mathbf{i} + 6(x^2 + y^2)\mathbf{j}$ is conservative using the mixed partials test, then find a potential function f such that $\mathbf{F} = \nabla f$.
2. Find a potential function for $\mathbf{F}(x, y, z) = (e^x \cos y + yz, xz - e^x \sin y, xy + z)$.
3. Find a potential function f for $\mathbf{F}(x, y, z) = 2xy\mathbf{i} + (x^2 - z^2)\mathbf{j} - 2yz\mathbf{k}$ and compute

$$\int_C \mathbf{F} \cdot d\mathbf{r}$$

where C is any path from $(0, 0, 0)$ to $(1, 2, 3)$.

4. Find a potential function f for

$$\mathbf{F}(x, y, z) = \left\langle \frac{1}{y}, -\frac{x}{y^2}, 2z - 1 \right\rangle$$

and use it to evaluate $\int_C \mathbf{F} \cdot d\mathbf{r}$ along the curve $C : \mathbf{r}(t) = \langle \sqrt{t}, t + 1, t^2 \rangle, 0 \leq t \leq 1$.

Applications

5. The namesake of “conservative” vector fields has roots in the fact that observable force fields are conservative, so as to obey the law of conservation of energy. Let’s verify this fact for the gravitational field in a 2-body system:

$$\mathbf{F}(x, y, z) = \frac{GMm}{(x^2 + y^2 + z^2)^{3/2}} \langle x, y, z \rangle$$

where M, m represent the masses of the two bodies and G is a physical constant. Use the curl test to show that $\mathbf{F}$ is conservative. Verify that the gravitational potential may be given by $V(x, y, z) = \frac{-GMm}{\sqrt{x^2 + y^2 + z^2}}$.

6. Your classmate in ECON 101 has recently gotten into the honey-selling business and needs to come up with a model describing how much it would cost to ship 10000L of honey from a factory to a destination. They suggest the following honey-transporting scheme: for some conservative vector field $\mathbf{F}$, the price of moving 10000L of honey from point A to B is about $\int_C \mathbf{F} \cdot \mathbf{T} \, ds$, where C is any curve linking A and B . Using your MATH 2551 expertise, what would you tell your classmate regarding the accuracy of their model?

Extensions

7. Let a, b, c, d, e be real numbers and

$$P(x, y, z) = 3x + 7y + 2z;$$

$$Q(x, y, z) = ax + by + 4z;$$

$$R(x, y, z) = cx + dy + ez.$$

For which values of the constants a, b, c, d, e is $\mathbf{F} = P\mathbf{i} + Q\mathbf{j} + R\mathbf{k}$ a conservative vector field?

8. This problem highlights an intricacy in the curl test. Consider the “vortex” vector field $\mathbf{F} = \left\langle \frac{-y}{x^2 + y^2}, \frac{x}{x^2 + y^2} \right\rangle$.

- (a) What is the domain of $\mathbf{F}$?
- (b) Show that $\mathbf{F}$ passes the curl test everywhere on its domain.
- (c) Show that

$$\int_C \mathbf{F} \cdot \mathbf{T} \, ds \neq 0$$

where C is the unit circle.

- (d) But $\mathbf{F}$ satisfies the curl test! By the fundamental theorem of line integrals, shouldn't the above integral be zero? What goes wrong?
- (e) Recompute the integral in (c) but replace C with the boundary of the square $[1, 2] \times [1, 2]$. What do you notice? Summarize your findings by filling in the blank below:

The integral $\int_C \mathbf{F} \cdot \mathbf{T} \, ds$ equals zero whenever the region bounded inside C does *not* contain _____.